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Liquidity Mining as an Attack Surface: Incentive-Induced Liquidity Attacks in Concentrated Liquidity Market Makers

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Abstract

Decentralized finance (DeFi) platforms commonly deploy volume-based incentive programs to bootstrap liquidity and improve market efficiency. We show that such subsidies, while well-intentioned, can create an underexplored economic attack surface in concentrated liquidity market makers (CLMMs). These subsidies enable a novel manipulation strategy. Attackers attract subsidized trading volume by providing highly concentrated liquidity, artificially inflating yields and luring victim liquidity providers into narrow price ranges. Once sufficient victim liquidity has accumulated, the attacker induces a price crash to extract value. We formalize this behavior as an incentive-induced liquidity attack and prove that it arises endogenously whenever subsidy intensity and volume caps exceed critical thresholds. Although subsidies are intended to bootstrap liquidity, these attacks ultimately lead to substantial liquidity depletion, causing platforms to incur incentive costs while market liquidity deteriorates. We conduct a large-scale measurement study on incentivized CLMM pools on BSC that are tied to Binance Alpha programs. We identify 29 attack events across 18 pools with 99 positive-loss victim addresses across 13 loss-bearing episodes and \$2.87M in measured victim loss. These measurements show that incentive programs can amplify manipulation by introducing externalities that enable new extraction opportunities. Based on our findings, we propose mitigation strategies that reshape incentive parameters and information disclosure to reduce liquidity-provider exposure to incentive-induced attacks. Our findings show that incentive mechanisms designed to improve market quality can instead create the security vulnerabilities they aim to prevent.

1 Introduction

To bootstrap liquidity and improve market efficiency, many decentralized finance (DeFi) platforms deploy volume-based

incentive programs that reward trading activity [25, 28, 53]. Among them, a large share target is decentralized exchanges (DEXs), a core infrastructure of DeFi that enables permissionless trading without centralized intermediaries [50, 55, 57]. Concentrated liquidity automated market makers (CLMMs) such as Uniswap v3 allow liquidity providers (LPs) to concentrate capital within specific price ranges rather than across the entire price curve [26, 27]. This mechanism improves capital efficiency while making liquidity distribution a key factor in price stability and execution quality [26, 31].

Across DeFi trading venues, bootstrapping campaigns often take the form of volume-based “points” programs. This form is widely adopted by DEXs such as Blast, Aerodrome, and PancakeSwap [3, 4, 35]. These programs provide external rewards for eligible trading activity, making participation privately attractive when the expected reward value exceeds trading costs. We refer to such volume-seeking participants as points farmers. In CLMMs, trades routed through an active price range generate fees for the LPs currently supplying liquidity in that range. As a result, subsidy-driven volume can raise apparent LP yields and attract additional liquidity into the pool [16, 22, 30].

We find that these incentive mechanisms, despite being designed to improve market efficiency, can create exploitable conditions in CLMMs [45]. As shown in Figure 1, attackers can create the appearance of deep liquidity, attracting both subsidized traders and victim liquidity providers. Specifically, attackers can exploit the properties of CLMMs by constructing ultra-concentrated liquidity positions around the prevailing price, thereby minimizing slippage and attracting a large share of points farmers. After sufficient liquidity accumulates, the attacker withdraws their position and executes a large trade that moves the price beyond the active range, leaving remaining LPs exposed to losses while the attacker exits with extracted value. This attack framing is consistent with growing attention to DeFi economic attacks and extractable value [18, 37, 41]. Rather than relying on a contract-level bug, such attacks can exploit public ordering, AMM price formation, flash-loan composability and DeFi-specific protocol semantics to extract

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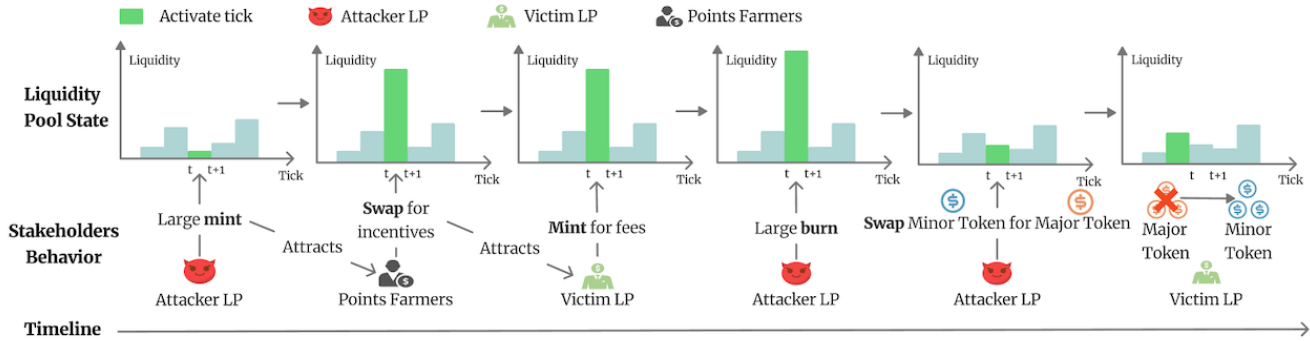


Figure 1: Timeline of an incentive exploitation attack in CLMMs. We label the two assets as the major token (higher market value, more widely traded) and the minor token (lower market value, less common and mainly traded in this pool).

value from other participants [42, 46, 52, 56].

To systematically study this incentive-induced economic attack surface in CLMMs, we combine theoretical analysis with empirical measurement. We first develop a formal model of CLMMs with volume-based incentives and show that when platform-provided incentives exceed critical thresholds, incentive-induced liquidity attacks arise endogenously and become increasingly severe as incentives grow. Importantly, although these attacks are predictable in equilibrium, rational LPs may still choose to supply liquidity ex ante, as the expected short-term yields dominate the anticipated risk of being attacked. Moreover, inflated yield induce herding behavior among myopic LPs, further amplifying victim participation and increasing the eventual damage of the attack. We then conduct a large-scale empirical measurement study of Uniswap v3-style pools on BNB Smart Chain over a ten-month period. We identify 29 attack incidents across 18 pools, with \$2.87M in measured victim loss borne by 99 positive-loss victim LPs across 13 loss-bearing episodes. These incidents are also accompanied by significant post-attack liquidity depletion at the pool level. Our theoretical and empirical analyses demonstrate that these attacks are not isolated anomalies but a systematic consequence of incentive design in CLMM-based platforms.

Beyond individual victim losses, these attacks undermine platform integrity at the systemic level. Specifically, incentive-induced liquidity attacks lead to two economic security consequences. First, platform incentives fail to generate genuine market activity and instead induce transient, subsidy-driven volume that leaves liquidity economically ineffective. Second, the capital of honest LPs is rendered unproductive and ultimately converted into devalued assets after the attack, resulting in direct losses to participants while the platform simultaneously expends incentives. Addressing these vulnerabilities requires principled changes to incentive and liquidity design.

Based on our theoretical and empirical findings, we propose a set of mitigation strategies that break different links in the attack chain. First, platforms can calibrate incentive caps and pool fees to keep markets out of the economically

profitable region for attackers. Second, platforms can delay reward settlement to blunt the timing advantage created by slow victim accumulation and fast extraction. Third, platforms can improve risk-relevant information disclosure by surfacing liquidity concentration and churn signals, which can reduce victim loading by making manipulation risk salient at entry. Finally, platforms can apply anomaly-triggered dynamic caps during prolonged wall-building phases to tighten incentives preemptively in at-risk pools.

2 Background

2.1 Mechanism of CLMM

Automated market makers (AMMs) are the primary liquidity mechanism in DEXes, where prices are determined algorithmically via on-chain liquidity pools [55]. CLMMs are a class of AMMs that extend earlier designs by allowing LPs to allocate capital over a specified price range $[P_L, P_U]$, instead of distributing liquidity uniformly across all prices.

CLMMs increase capital efficiency by concentrating liquidity near the prevailing market price: for a given amount of capital, more liquidity is available where trades are most likely to occur, resulting in lower slippage and higher fee generation when trading activity is concentrated within the selected range.

CLMMs discretize the price space into a sequence of *ticks*, where each tick corresponds to a fixed price increment (e.g., 0.01%). Let P denote the spot price of one pool asset in units of the other, and let $z = \sqrt{P}$ denote the corresponding square-root price. LPs specify their positions using tick boundaries, equivalently denoted as $[z_L, z_U]$ in square-root-price space. Within the specified range, the pool behaves as a constant-product AMMs; once the market price moves outside the range, the liquidity position becomes inactive and holds only one of the two assets.

Similar to traditional AMMs, LPs in CLMMs earn fees proportional to their share of active liquidity. When a swap

occurs, a fixed percentage fee is collected and distributed pro rata among LPs whose liquidity is active at the prevailing market price. Once the price leaves the interval over which an LP has provided liquidity, the position ceases to earn fees until the price re-enters the range.

2.2 Liquidity Incentive Programs

Liquidity incentive programs are widely used in DeFi to bootstrap activity for selected tokens and pools. A sponsor, such as a token issuer, DEX platform, or ecosystem allocator, provides external rewards to encourage users to supply liquidity or generate trading volume on a designated venue. These rewards are distributed according to public rules over a measurement window, such as an hourly, daily, or weekly window. Volume-linked programs are especially relevant to our setting: they subsidize trading activity and can make participation privately attractive even when organic demand is limited. As trading activity increases, LPs may observe higher fee yields and supply additional liquidity, creating a feedback loop between subsidized volume and liquidity entry.

A representative example is **Binance Alpha Points** [2], a points-based campaign centered on a curated set of “Alpha tokens” [1]. Users accumulate points by interacting with eligible tokens and can use points to qualify for early-access opportunities such as airdrops or token distribution events. These rewards are external to the CLMM itself: they are provided by the platform to promote user activity, rather than being paid out from the pool’s swap fees. Thus, the incentive program is a non-zero-sum external subsidy layered on top of the trading venue. Points may not be immediately cash-settled, but users can assign them an expected value because they grant access to future token allocations or campaign benefits. In this study, we use Alpha-Points-style programs as an example and study how such volume-linked external incentives interact with CLMM liquidity to create incentive-induced liquidity attacks.

3 Related Work

3.1 DeFi Attacks

Since DeFi has become a core component of the blockchain ecosystem, attacks targeting DeFi protocols have attracted significant attention [39, 44, 44, 49, 57]. Existing studies on DeFi attacks can be broadly categorized into two classes: vulnerabilities in *protocol implementations* and exploits arising from *economic mechanism design*.

Recent researches focus on vulnerabilities in smart contracts, including logic errors, reentrancy, and improper state management [7, 13, 24, 38, 43, 51]. Beyond smart contracts, researchers have also investigated vulnerabilities in auxiliary components of the DeFi ecosystem, such as wallets [19, 20] and off-chain oracles [8, 29].

Researches also explored attacks that exploit economic mechanisms and market structures in DeFi systems, rather than conventional implementation bugs. A central example is Miner or Maximal Extractable Value (MEV) [18, 41], where adversaries profit by strategically ordering, inserting, or suppressing transactions [9–11]. Although these attacks do not necessarily violate protocol rules or exploit code-level vulnerabilities, prior work has shown that they can systematically harm users and market quality through value extraction, price distortion, and degraded execution.

These studies show that DeFi attacks can emerge not only from technical vulnerabilities, but also from economic mechanisms that adversaries can exploit. However, existing work primarily focuses on transaction ordering, temporary capital amplification, oracle dependence, liquidation mechanics, and cross-market price discrepancies. While incentive programs are now widely used to bootstrap DeFi markets, how incentive programs interaction with concentrated liquidity remains unclear from an adversarial perspective. Our work aims to explore how volume-based incentives interact with CLMM liquidity to manufacture order flow, attract LP liquidity, and enable profitable unwinds.

3.2 Price Manipulation in Financial Markets

Price manipulation has long been studied in traditional financial markets as a means of extracting profit through targeted attacks and arbitrage opportunities. A common pattern across manipulation strategies is the combination of *price pinning* and *sudden, drastic price movements*, which enables manipulators to trade against distorted prices.

Prior work has examined several mechanisms through which manipulation can arise, including market power, information asymmetry, and coordinated trading. Foundational theoretical work shows that manipulators can corner the market by accumulating dominant inventory positions or otherwise exploiting market power to distort prices [17, 32]. Information-based manipulation instead leverages asymmetric beliefs or imprecise information; for example, rumors can create buying pressure that benefits the manipulator [47]. Empirical studies further show that manipulation is prevalent and profitable in practice [5, 14], including pump-and-dump schemes orchestrated by colluding brokers in emerging markets [34]. Across these mechanisms, manipulators profit by inducing other market participants to trade at distorted prices.

In cryptocurrency markets, price manipulation techniques share many similarities with traditional markets but also arise through blockchain-specific mechanisms. Pump-and-dump schemes, often coordinated via social media, have been extensively documented [21, 33, 36, 54]. Prior work also documents artificial activity in blockchain markets, including wash trading on decentralized exchanges, fake volume on centralized exchanges, and wash trading in NFT markets [15, 40, 48]. Studies about token incentives and airdrops show that exter-

nal rewards can induce strategic participation and reward-seeking behavior [6, 23]. In DeFi, manipulation can further exploit protocol-level dependencies, such as oracle attacks based on temporary DEX price distortions [42, 52, 57], as well as liquidity-pool abuse involving rug pulls, sniper bots, and wash-trading activity [12].

These studies show that price manipulation can arise through multiple channels, including market power, information asymmetry, coordinated trading, and protocol-level price dependencies. However, existing work primarily focuses on how adversaries distort prices through direct trading behavior, misleading information, or oracle-facing price movements. How external incentive programs affect price stability and manipulation risk in concentrated-liquidity markets remains underexplored. Our work addresses this gap by studying incentive-induced manipulation in CLMMs, where volume-based rewards interact with localized liquidity.

4 Incentive-Induced Liquidity Attacks

We first formalize an abstract model of a CLMM and a trading-incentive program, capturing the essential mechanics relevant to incentive-induced liquidity attacks. We then define the threat model. The central mechanism operates as follows. An attacker holding inventory of the alpha token strategically provides concentrated liquidity, thereby reducing execution costs and attracting incentive-driven traders. The resulting trading volume generates fee income, which in turn attracts victim LPs seeking those returns. Once sufficient victim liquidity accumulates, the attacker withdraws and sells inventory through the victims' positions, extracting value at their expense. All formal proofs are provided in Appendix A.

4.1 CLMM Model and Trading Incentives

CLMM. We consider a CLMM trading token0 (the *minor* token) and token1 (the *major* token). Let P denote the spot price, defined as the amount of token1 per unit token0. We equivalently write $P = z^2$, where z is the square-root price. Let $z_0 = \sqrt{P_0}$ denote the initial square-root price.

The price space is discretized into ticks. When P lies within a tick interval, liquidity within that interval remains constant, and trades move the price according to a fixed liquidity parameter L . Within a single tick, price evolution follows the standard CLMM relations:

$$\Delta x = L \left(\frac{1}{\sqrt{P_b}} - \frac{1}{\sqrt{P_a}} \right), \quad \Delta y = L(\sqrt{P_b} - \sqrt{P_a}),$$

where P_b and P_a are the prices before and after a swap, and $(\Delta x, \Delta y)$ are the corresponding token balance changes.

LPs. There are $N > 1$ LPs. Each LP k supplies liquidity L_k over a chosen price interval $[P_L, P_U]$. When the market

price inside this range, the LP's liquidity is *active* and earns fees proportional to trading volume; otherwise the position is *inactive*.

An LP may withdraw liquidity at any time. We denote LP k 's net payoff in token1 units at price P as

$$\Pi_k(P) = u_k(P) - \text{IL}_k(P),$$

where $u_k(P)$ is the fee income earned while the position is active, and $\text{IL}_k(P)$ captures range-dependent exposure to price changes (impermanent loss relative to holding).

Fees. Each swap incurs a proportional fee at rate f , distributed among all active liquidity. Let L^{active} denote the total active liquidity at the current price. Abstracting away implementation details, fee accrual per unit of liquidity satisfies

$$\text{fee per unit liquidity} \propto \frac{\text{executed trading volume}}{L^{\text{active}}}.$$

Thus, for fixed volume, additional active liquidity dilutes the fee income earned per unit of liquidity.

Trading-incentive program. In addition to swap fees, we consider an incentive program that rewards traders for generating trading volume in the pool. Let $\sigma \in [0, 1]$ denote the platform subsidy (calculated in token1 units) paid to traders on gross trading volume up to a cap Q (volume beyond Q earns no additional rewards). Here σ should be interpreted as the total cash-equivalent incentive intensity, potentially including the discounted value of token/point rewards. The cap Q abstracts the reward-relevant volume range within a measurement window: beyond this range, additional volume no longer materially increases the trader's expected reward.

Incentive-driven traders and volume V . We model a population of *incentive-driven traders* who are not arbitraging price discrepancies. Instead, they trade to harvest incentive benefits. For gross volume v executed within the measurement window, a representative trader's net benefit is

$$U_T(v) = \sigma \cdot \min\{v, Q\} - f \cdot v - \text{MC}(v),$$

where f is the pool fee rate and $\text{MC}(v)$ captures expected execution costs (e.g., price impact and other frictions). Ignoring the cap, an interior optimum satisfies

$$\sigma - f - \text{MC}'(V) = 0. \quad (1)$$

We denote the trader-optimal volume by

$$V^*(\sigma) := \arg \max_{v \geq 0} U_T(v),$$

and use V for the realized aggregate executed volume within the window. The cap is binding when aggregate reward-relevant volume reaches $V = Q$.

4.2 Threat Model

Participants. We consider three types of participants:

Attacker. A participant who holds an initial inventory of token0 and seeks to convert it to token1. The attacker supplies concentrated liquidity L_A to shape market conditions, and subsequently sells inventory through accumulated liquidity.

Victim LPs. Non-adversarial LPs who observe elevated fee yields and supply liquidity L_V to capture returns. Critically, victim LPs do not distinguish between organic trading activity and volume induced by incentive programs—they respond to yields regardless of their source.

Incentive-driven traders. Traders who execute volume solely to harvest incentive rewards. They are indifferent to the fundamental value of the alpha token and price movements, while their payoff depends only on executed volume relative to execution costs. This indifference is central: incentive-driven traders generate volume in any pool where subsidies exceed costs, regardless of whether that volume benefits or harms other participants.

Attacker objective. The attacker’s profit in token1 units is

$$\Pi_A = \underbrace{f \cdot V \cdot \frac{L_A}{L_A + L_V}}_{\text{Fee capture}} + \underbrace{U_{\text{dump}}}_{\text{Crash extraction}} - \underbrace{c_A \cdot X_{\text{sold}}}_{\text{Inventory cost}} - \kappa, \quad (2)$$

where V is the aggregate gross trading volume realized in the pool during the window, U_{dump} is the value extracted by selling through residual liquidity, X_{sold} is the total token0 sold by the attacker, c_A is the attacker’s unit inventory cost or reservation value for token0, and κ captures frictions such as gas and capital lockup.

LP Victim loss. Victim LPs face a corresponding participation tradeoff. They earn fee income from the manufactured volume, but they also bear an effective per-unit cost c_V . This term summarizes ordinary liquidity-provision costs, expected impermanent-loss risk, and expected crash exposure. In the simplified one-tick model, the victim-side payoff from supplying liquidity L_V is

$$\Pi_V(L_V) = \underbrace{fQ \cdot \frac{L_V}{L_A + L_V}}_{\text{Fee income}} - \underbrace{c_V L_V}_{\text{Participation cost and crash exposure}}.$$

Here, the crash-exposure component of c_V corresponds to unrecovered non-alpha value after the breach, net of the residual post-breach value of alpha tokens retained in, or recovered from, the LP position.

It should be noted that attacker profit and victim loss are related but need not match. Attacker profit includes fee capture and extraction through all liquidity crossed during the selloff, including adjacent or downstream ticks outside our identified victim set. Victim loss measures only the realized

crash exposure of identified victim LPs that retain one-tick liquidity. Therefore, attacker profit and victim loss are distinct economic quantities rather than a zero-sum pair.

Attacker capabilities. We assume the attacker can: supply and withdraw liquidity (choose L_A and the attacked range) at will; trade in the pool (including selling inventory X_{sold}); participate in the incentive program under its public rules (parameterized by σ and Q); and influence realized volume V indirectly by shaping market depth (via L_A), which reduces traders’ execution costs $\text{MC}(\cdot)$ under the incentive program.

Attack surface. Consistent with our empirical focus and the cleanest observed configuration, we analyze attacks concentrated on a single tick interval containing the current price z_0 . Both attacker and victims provide liquidity on a narrow range $[z_L, z_U]$ containing z_0 . We assume the program’s effective subsidized volume fits within this tick under attacker liquidity:

$$Q \leq V_{\text{max}}^{\text{tick}}(L_A) \equiv \frac{L_A}{1-f}(z_U - z_0). \quad (3)$$

This ensures the relevant trading dynamics remain within the attacked tick even without victim liquidity.

Attack overview. An attack episode unfolds sequentially, with each stage creating the conditions for the next:

Wall-building: The attacker deposits concentrated liquidity L_A on $[z_L, z_U]$, creating deep market depth that reduces execution costs for traders.

Volume generation: The reduced execution costs attract incentive-driven traders, who generate volume V in response to subsidy σ . This trading activity produces fee income that elevates the pool’s yields.

Victim entry: Victim LPs observe the elevated yields and supply liquidity L_V , unaware that the volume is manufactured by incentive programs rather than organic demand.

Unwind and extraction: Once victim liquidity L_V is sufficient to absorb the attacker’s inventory, the attacker withdraws L_A and sells token0 through the remaining liquidity. The resulting price decline leaves victims holding depreciated assets while the attacker exits with token1.

4.3 Equilibrium Analysis

4.3.1 Analytical Regime

We use two assumptions to obtain a closed-form equilibrium: the incentive cap binds, and trading remains within the attacked tick. Together, they fix subsidized volume at Q and isolate the empirical one-tick wall-and-breach mechanism.

Assumption 1 (Cap-Binding Regime). *The subsidy satisfies $\sigma \geq \sigma_Q \equiv \sigma_{\text{cap}}(Q; L_A)$, where*

$$\sigma_{\text{cap}}(Q; L) \equiv 1 - \frac{P_e(1-f)}{\left(z_0 + \frac{(1-f)Q}{L}\right)^2}.$$

Under this condition the program cap binds and $V^(\sigma; L_A + L_V) = Q$ for any $L_V \geq 0$.*

Assumption 2 (Tick Feasibility). *The volume cap satisfies $Q \leq V_{\text{max}}^{\text{tick}}(L_A) \equiv \frac{L_A}{1-f}(z_U - z_0)$, ensuring all trading stays within the one-tick range regardless of victim liquidity $L_V \geq 0$.*

These assumptions imply that equilibrium volume $V^* = Q$ is *exogenous* to the victim's liquidity decision, breaking the fixed-point problem that would otherwise arise.

4.3.2 Equilibrium Characterization

We analyze a sequential game among an attacker, victim LPs, and incentive-driven traders. The equilibrium outcome follows a feedback loop: trading incentives manufacture predictable order flow, the resulting fee income attracts liquidity into a narrow CLMM range, and the accumulated victim liquidity creates an exploitable surface for the attacker. We now formalize each step of this mechanism. All proofs are collected in Appendix A.

We first analyze the trading side of the mechanism: when incentives induce participation and how much volume they generate. We use σ_0 to denote the participation threshold below which incentive-driven traders do not trade. For a given volume cap Q and active liquidity L_0 , $\sigma_{\text{cap}}(Q; L_0)$ denotes the subsidy level at which the volume cap binds. We write $\sigma_Q \equiv \sigma_{\text{cap}}(Q; L_A)$ for the cap-binding threshold under attacker liquidity alone.

Theorem 1 (Subsidies Manufacture Volume). *Under the one-tick assumption with active liquidity $L_0 = L_A + L_V > 0$, the equilibrium trading volume $V^*(\sigma; L_0)$ exhibits three regimes as the subsidy rate σ increases:*

1. No participation: $V^*(\sigma; L_0) = 0$ for $\sigma \leq \sigma_0$;
2. Monotone growth: $V^*(\sigma; L_0)$ is nondecreasing in σ for $\sigma > \sigma_0$;
3. Cap binding: for $\sigma \geq \sigma_{\text{cap}}(Q; L_0)$, the volume is pinned at the program cap, $V^*(\sigma; L_0) = Q$.

The theorem follows the next lemma, which provides the closed-form characterization used throughout the rest parts.

Lemma 2 (Closed-Form Volume Under One-Tick). *Under the conditions of Theorem 1, for $\sigma > \sigma_0$ the uncapped optimum is*

$$\tilde{V}(\sigma; L_0) = \frac{L_0}{1-f} \left(\sqrt{\frac{P_e(1-f)}{1-\sigma}} - z_0 \right), \quad (4)$$

and equilibrium volume is $V^(\sigma; L_0) = \min\{Q, \tilde{V}(\sigma; L_0)\}$. Moreover, $V^*(\sigma; L_0) = Q$ whenever $\sigma \geq \sigma_{\text{cap}}(Q; L_0)$ (in particular whenever $\sigma \geq \sigma_Q$).*

This closed-form expression makes the first step of the mechanism explicit:

Takeaway 1 (Thresholded volume manufacturing). Theorem 1 and Lemma 2 show that incentive-driven trading has a threshold structure. Below the participation threshold, traders do not enter; above it, trading volume grows predictably with incentive strength and eventually becomes cap-bound.

Operationally, since program parameters (σ, Q) are public, Theorem 1 implies that the attacker can predict the magnitude of induced volume for any given liquidity level. Given predictable order flow, we next ask how victims respond.

Theorem 3 (Cap-Induced Volume Loads Liquidity). *Assume $\sigma \geq \sigma_Q$. Fix attacker liquidity $L_A > 0$ and an effective cost $c_V > 0$ for victim liquidity supply. Then victim liquidity in equilibrium is increasing in Q and decreasing in c_V . In particular, there exist parameter regimes where $L_V^*(Q) > 0$ even when victims are fully rational and correctly price manipulation risk into c_V .*

The theorem is implied by the following technical best-response lemma.

Lemma 4 (Victim Best Response). *Under $\sigma \geq \sigma_Q$, the victim's optimal liquidity is*

$$L_V^*(Q) = \max \left\{ 0, \sqrt{\frac{fQL_A}{c_V}} - L_A \right\}. \quad (5)$$

At an interior optimum, total active liquidity satisfies $L_A + L_V^ = \sqrt{fQL_A/c_V}$.*

Lemma 4 makes the liquidity-loading channel explicit.

Takeaway 2 (Liquidity loading). Theorem 3 and Lemma 4 show that cap-induced volume can load victim liquidity. Once incentives generate sufficient trading volume, the resulting fee pool makes liquidity provision privately attractive, even for rational LPs that price manipulation risk into their effective cost.

This gives rise to the “rational victim paradox”: even when an LP anticipates crash risk through c_V , sufficiently large cap-induced volume Q can still make entry privately attractive. If victims additionally underweight crash risk, their effective cost is smaller, and Lemma 4 implies strictly larger liquidity supply, further amplifying the attack surface. In practice, this channel can be reinforced by interfaces that highlight trailing fee yields or campaign metrics, which mechanically track manufactured volume.

We next characterize when the attacker prefers launching the attack over an outside option such as direct token0 sale.

Lemma 5 (Dump-to-Floor). *Starting from square-root price z_1 , the attacker optimally sells until the price reaches*

$$z_2^* = \max\{z_{\text{floor}}, z_{\bar{A}}, z^\dagger\} \quad \text{subject to } z_2^* \leq z_1,$$

where z_{floor} is liquidity floor; $z_{\bar{A}}$ is inventory-implied floor; and $z^\dagger = \sqrt{c_A/(1-f)}$ is marginal-cost stopping point.

Using this, we obtain the high-level feasibility result: profitable attacks exist if and only if the cap exceeds a threshold.

Theorem 6 (Attack Feasibility Threshold). *Assume $\sigma \geq \sigma_Q$. There exists a volume-cap threshold Q^* such that the attack is profitable if and only if $Q \geq Q^*$. Formally, let Π_A^Q denote the attacker payoff and Π_{direct} the outside-option payoff from direct token0 sale. Then Π_A^Q is nondecreasing in Q , and*

$$Q^* \equiv \inf\{Q \in [0, V_{\max}^{\text{tick}}(L_A)] : \Pi_A^Q \geq \Pi_{\text{direct}}\}, \quad (6)$$

with $Q^* = +\infty$ if the set is empty (in which case the attack is never profitable).

The threshold Q^* has a direct economic interpretation: it is the smallest effective cap at which launching the attack weakly dominates the outside option of directly selling token0. Thus, once subsidies are large enough to make the cap bind, attack feasibility becomes a design boundary governed by (σ, Q) and the market's cost parameters.

Takeaway 3 (Attack feasibility threshold). Lemma 5 and Theorem 6 show that once liquidity accumulates, attack profitability is governed by the effective volume cap. Below Q^* , the attacker prefers the outside option; above Q^* , the withdrawal-and-dump strategy becomes profitable in equilibrium.

Together, Takeaways 1–3 imply a fundamental volume–security tension: strong incentives are effective at manufacturing volume, but they also load liquidity and enlarge the crash surface until attacks become profitable. We summarize this tension as follows.

Corollary 7 (Volume–Security Tradeoff). *A platform that wants to attract volume $\underline{V}$ must offer subsidies large enough to generate that volume, but doing so may cross the attack threshold. The optimal safe subsidy is*

$$\sigma^{\text{opt}} = \min\{\sigma : V^*(\sigma) \geq \underline{V}\} \quad \text{subject to } \sigma < \sigma_Q \text{ or } Q < Q^*. \quad (7)$$

If achieving $\underline{V}$ requires $\sigma \geq \sigma_Q$ and $Q \geq Q^$, the platform faces a binding tradeoff: either lower the volume target or accept attack risk.*

In other words, the very subsidies designed to bootstrap a healthy market can, if set too aggressively, turn that market into an attack surface. The platform's design space therefore splits into three regimes:

Safe bootstrapping ($\sigma < \sigma_Q$ or $Q < Q^*$): Subsidies attract volume without crossing the attack threshold. LPs earn fees and the platform grows.

Vulnerable growth ($\sigma \geq \sigma_Q$ and $Q \geq Q^*$): The platform achieves volume target, but attacks become profitable. LP losses are a consequence of the incentive design.

Constrained optimum: Platform sets σ below the attack threshold, accepting reduced volume to protect LPs.

5 Methods

This section describes how we identify and characterize one-tick liquidity attacks in CLMM on BNB Smart Chain. We combine on-chain event traces with an auxiliary off-chain price signal with four steps: we scope the relevant pool universe, detect candidate liquidity-wall episodes from Mint/Burn activity on one-tick ranges, validate candidates by requiring sharp post-withdrawal price declines realized within the same pool, and replay tick-level liquidity to infer participant roles and cluster related addresses.

5.1 Data Sources

On-chain traces. We enumerate all Uniswap-v3-style pools on BSC by scanning the factory PoolCreated logs, which yield each pool address and token pair. We then collect decoded on-chain logs (Mint, Burn, Swap) from Allium for January 1, 2025 to October 1, 2025. Mint/Burn events provide tick ranges and signed liquidity deltas, enabling us to reconstruct tick-localized liquidity changes; Swap events provide pool-internal price evolution for validation.

Off-chain price signal. To efficiently detect sharp price drops (without exhaustively scanning all Swap logs across all pools), we additionally use an hourly DEX price series from the Allium API as a coarse filter. We only fall back to Swap-level validation for candidates that exhibit a corresponding sharp drop in this price signal.

Alpha-token pools. We restrict the analysis to pools tied to active incentive programs. We collect the Binance Alpha token recoded on September 16, 2025 and extract 229 token contract addresses.¹ We retain only pools that contain at least one Alpha token. This scoping matches our threat model: incentive programs concentrate activity on a subset of tokens, and attackers target the pools that receive this directed activity.

5.2 Detecting and Validating Episodes

Candidate identification. We identify candidate attack episodes by detecting abnormal liquidity activity localized to a single tick interval (positions with tickUpper = tickLower + 1). For each pool and one-tick interval, we aggregate Mint and

¹<https://www.binance.com/bapi/defi/v1/public/alpha-trade/aggTicker24?dataType=aggregate>

Burn events into a daily net liquidity-change series and detect anomalous days using a per-(pool, tick) outlier rule based on three-standard-deviation thresholds on absolute change.

We consolidate consecutive anomalous days with the same direction into "build-wall" groups (Mint-dominated) and "remove-wall" groups (Burn-dominated). We then pair each build-wall group with the nearest subsequent remove-wall group on the same pool and tick interval, yielding candidate episodes defined by the pool address, attacked tick interval, and build and removal periods.

Validation. We do not treat every large liquidity withdrawal as an attack. According to the attack features, we require a sharp price decline that occurs shortly after the removal signal and is attributable to swaps *within the same pool*.

We validate candidates in two stages. First, we scan the hourly Alpha-token price series to identify declines of at least 10% within two hours following the candidate removal window. Second, for each retained candidate, we load Swap events from the same pool in a narrow window around the decline and verify that the pool tick moves in the direction consistent with an Alpha-token price drop under the pool's token ordering. This two-stage validation filters out cross-venue noise (e.g., price movements driven by activity in other pools) and confirms that the crash is realized through on-chain swaps in the attacked pool.

5.3 Role Inference and Measurement Setup

Event-anchored replay. We characterize each confirmed episode using four anchor times that demarcate the attack lifecycle in chronological order:

t_m : the onset of wall construction, marked by the first Mint into the attacked one-tick range;

t_0 : the critical withdrawal time, when the wall begins to be dismantled;

t_{oot} : the first Swap that moves the price out of the attacked tick, which we use as a pool-internal alignment check;

t_b : the breach time, when the price drops by at least 10% from the attacked tick.

We identify the withdrawal anchor t_0 as the largest Burn on the attacked range within a short window preceding t_b . We then replay Mint and Burn events in on-chain order to reconstruct each address's effective one-tick liquidity over $[t_m, t_b]$ and compute its share of the wall immediately prior to withdrawal.

Wall concentration. To identify who controls the liquidity wall, we compute each address's time-weighted liquidity share over $[t_m, t_b]$. Specifically, we integrate per-address one-tick liquidity $L_u(t)$ over time and normalize it by total one-tick liquidity in the episode, so that address shares sum to one within each episode. We use this concentration profile both to measure whether the wall is controlled by a small number of entities and to support role inference.

Role labeling. We assign each address to a role using the replayed liquidity trajectories and the identities of Swap initiators during $[t_0, t_b]$. Because the exact moment of breach can be defined either by the first pool-internal out-of-tick swap or by the hourly 10% breach time used in detection, we carry both end-time conventions forward: $t_{end} \in \{t_{oot}, t_b\}$. Within each crash window $[t_0, t_{end}]$, we label a *pressure swap* as a swap that moves the price in the breach direction and extracts the non-alpha token.

Using observable on-chain actions, we assign addresses to the following roles:

Victims: addresses that retain positive one-tick liquidity at the chosen end time, i.e., $L_u(t_{end}) > 0$, and are therefore exposed through the crash;

Wall providers: addresses that both mint and burn one-tick liquidity within $[t_m, t_{end}]$, capturing wall construction and dismantling;

Pressure traders: addresses that initiate at least one pressure swap within $[t_0, t_{end}]$, capturing directional execution into residual liquidity.

We classify an address as an *attacker* if it satisfies both the wall-provider and pressure-trader criteria under the same end-time convention. We then designate the address with the largest pre-withdrawal wall share as the *main attacker* for the episode. This construction maps observable behavior to the model's key roles: victims retain liquidity through the crash, while attackers both control the wall and exert crash-period directional pressure.

Victim loss measurement. We measure victim loss as the realized crash-exposure component of c_V : unrecovered non-alpha value net of the residual post-breach value of alpha tokens. For each victim LP u , we reconstruct tokens deposited through Mints, tokens recovered through Burns before t_b , and tokens still remaining in the LP position at t_b .

We define E_u as the realized crash exposure: the non-alpha value the LP would retain under a hold counterfactual, minus the non-alpha assets actually recovered or retained through the LP position, offset by the post-breach value of residual alpha tokens. Residual alpha tokens are marked using the volume-weighted average alpha price over swaps in $[t_b, t_b + 1h]$. The measured victim loss is $\text{Loss}_u = \max\{0, E_u\}$.

Attacker profit measurement. We measure attacker profit separately from victim loss. For each identified attacker address, we aggregate its net non-alpha cash flow from breach-direction swaps during the crash window $[t_0, t_b]$, and sum these cash flows across attacker addresses to obtain event-level profit.

6 Experimental Results

Applying the detection pipeline described in Section 5, we identify 29 one-tick attack episodes across 18 pools. Appendix B reports the full filtering breakdown from the initial Alpha-token pool universe to the final validated event set. To

analyze individual-level behavior in incentive-induced liquidity attacks, we decompose each episode into a sequence of stages: wall construction, liquidity attraction, wall dismantling, price breach, and post-breach degradation.

Our empirical analysis proceeds along three dimensions. First, we characterize the temporal dynamics of attack episodes, measuring the duration of each stage and the relative response speeds of different stakeholders. Second, we quantify the market impact of these episodes, including price movements, trading-volume amplification, and changes in active liquidity. Finally, we examine participant-level behavior within each episode, focusing on how attackers, liquidity providers, and traders adopt distinct strategies over the course of the attack.

6.1 Attack Dynamics and Temporal Structure

We analyze the temporal structure of the 29 one-tick episodes to test the timing implication of Takeaway 2: attackers must keep the wall in place long enough to attract victim liquidity, but can execute extraction rapidly once they begin dismantling the wall. We proceed in two steps. First, we measure coarse phase gaps from wall construction to withdrawal and breach to quantify the preparation–extraction asymmetry. Second, we zoom in on wall formation and dismantling by introducing additional anchors that separate ramp-up, maintenance, and the onset of withdrawal.

6.1.1 Phase Gaps

We test the preparation-extraction asymmetry by measuring three time gaps for each episode. Table 3 in Appendix C reports the binned distributions of these temporal gaps.

t_0 : the first post-peak drop below 95% of the peak, marking the onset of dismantling.

$\Delta_{\text{build}} = t_0 - t_m$: we measure how long the attacker maintains the liquidity wall, from the first wall mint to the onset of wall dismantling.

$\Delta_{\text{extract}} = t_b - t_0$: we measure how quickly the crash unfolds once dismantling begins, from the withdrawal anchor to the 10% price breach.

$\Delta_{\text{oout}} = t_b - t_{\text{oout}}$: we measure the alignment between the out-of-tick event and the hourly 10% breach used in detection; smaller values indicate tighter alignment.

We observe a sharp asymmetry between preparation and execution. Attackers often prepare slowly: in 12 of 29 episodes, Δ_{build} exceeds three days, indicating that attackers maintain the wall long enough for victim LPs to observe elevated returns and enter. Attackers then execute quickly: in 18 of 29 episodes, Δ_{extract} falls within 24 hours, so the price breaches before most victims can react. This slow build-up followed by fast extraction matches the operational logic of the attack: attackers wait to accumulate victim liquidity, then withdraw and extract before victims exit.

We also validate measurement quality using Δ_{oout} . In eight episodes, we find $\Delta_{\text{oout}} \leq 30$ minutes, which indicates that the hourly breach timestamp t_b closely tracks the pool-internal moment when the price leaves the attacked tick.

6.1.2 Wall Formation and Dismantling

We refine the wall timeline by introducing additional anchors that separate ramp-up, maintenance, and onset of dismantling:

t_{50} : the first time one-tick liquidity reaches 50% of eventual peak, marking if the wall becomes substantial.

t_{peak} : the time of peak liquidity on the attacked tick, capturing the end of wall buildup.

The binned distributions of the refined wall-timing gaps are reported in the [Online Appendix](#).

We find that wall formation speed varies substantially across episodes. In seven episodes, attackers reach 50% of peak liquidity at the same time as the initial mint, consistent with a single large deposit. In four episodes, attackers take more than three days to reach 50%, consistent with incremental deployment over time.

We also find that attackers often maintain the wall for longer than they build it: in eight episodes, $t_{\text{peak}} - t_{50} > 3$ days, consistent with repeated reinforcement or long-lived positions that continue to attract victims. After the wall peaks, attackers dismantle it rapidly: in 19 of 29 episodes, $t_0 - t_{\text{peak}}$ falls within one hour. This sharp transition from slow maintenance to fast dismantling matches the model: attackers keep the wall present to load victim liquidity (Theorem 3), then withdraw quickly ahead of the ensuing price move (Lemma 5). The clean separation also supports using t_0 as a stable boundary for stage-aligned analysis.

6.2 Market Impact of Attacks

We quantify how attacks reshape observable market conditions, connecting stage-level market signals to Takeaways 1 and 3. To make episodes comparable despite heterogeneous durations, we align episode to key anchors and summarize market observables in four stage windows with $W = 24$ hours: *Before* = $[t_m - W, t_m)$, *Prep* = $[t_m, t_0)$, *Attack* = $[t_0, t_b]$, and *Post* = $(t_b, t_b + W]$. We then test the model’s stage-specific predictions using complementary signals. We first measure *volume lift* to check whether the wall attracts incentive-driven order flow. We next examine *realized volatility* and *price displacement* to distinguish price pinning during preparation from downward repricing during execution. Finally, we measure *liquidity degradation* to verify that attackers remove the wall before (or at) the breach, leaving the tick depleted.

6.2.1 Volume Lift

We start with volume because it is the direct channel through which incentives and the wall interact. Takeaway 1 implies

that subsidies can manufacture volume, so if the wall successfully attracts incentive-driven trading, we should observe higher swap activity during *Prep* and potentially bursty activity during *Attack*. We therefore compute swap volume at hourly granularity using the absolute non-alpha token flow. We treat the *Before* window as the baseline and denote its mean volume by μ_{pre} . For each subsequent stage s , we compute the *volume lift* as $Lift_s = \mu_s / \max(\mu_{\text{pre}}, \epsilon)$, where ϵ prevents unstable ratios when the baseline is near zero.

We summarize three volume-lift metrics: $Lift_{\text{prep}}$ for wall construction $[t_m, t_0)$, $Lift_{\text{atk}}$ for execution $[t_0, t_b]$, and $Lift_{\text{post}}$ for the post-breach window $(t_b, t_b + W]$. Table 4 in Appendix C reports the corresponding binned distributions.

We confirm all three predictions. During preparation, 20 of 29 episodes exhibit above-baseline volume ($Lift_{\text{prep}} \geq 1 \times$), and most increases remain modest (11 fall in the $1\text{--}1.5 \times$ bucket), indicating that the wall attracts additional trading without dramatic spikes. During execution, volume becomes more dispersed: 17 of 29 episodes exceed baseline, and 4 exceed $5 \times$, consistent with concentrated bursts as the attacker extracts. After the breach, activity collapses: 23 of 29 episodes fall below baseline ($Lift_{\text{post}} < 1 \times$), which shows that the elevated volume does not persist once the wall disappears.

6.2.2 Price Volatility

After we establish that the wall lifts volume, we ask how the wall reshapes *price dynamics* while trading occurs. A one-tick wall should suppress price impact during *Prep* by concentrating depth near the active tick, but volatility should rise after wall removal because the same directional flow traverses a thinner curve. To measure this effect directly from on-chain activity, we construct realized volatility from swap-induced tick movements. For adjacent swaps, we compute the log return as $r_i = \ln(1.0001) \cdot (\text{tick}_i - \text{tick}_{i-1})$ and define stage volatility as $RV = \sqrt{\sum_i r_i^2}$.

Figure 2 summarizes RV across the four stages. We expect *Prep* to be the quietest stage, *Attack* to remain moderate until the wall is mostly gone, and *Post* to exhibit the largest volatility once the market reprices after the breach. The data match these predictions. *Prep* is the quietest stage (median RV = **0.0119**), followed by *Attack* (**0.1593**), and both lie below the *Before* baseline (**0.2532**). After the breach, *Post* volatility jumps to a median of **0.8843** with a substantially heavier upper tail, consistent with rapid repricing once the wall no longer absorbs flow.

6.2.3 Price Displacement

Volatility captures fluctuations, but we also want to capture the *directional* repricing that distinguishes wall pinning from crash execution. The model predicts small displacement around wall formation (the wall stabilizes price while volume is manufactured) and a sharp downward displacement

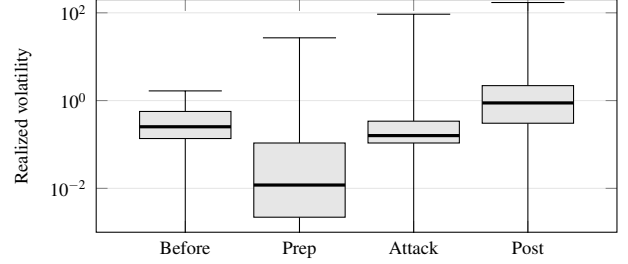


Figure 2: Box-and-whisker summary of realized volatility across four stages. Each box uses the 25th, 50th, and 75th percentiles and whiskers span the min and max. We use a log-scaled y-axis, and values below 10^{-4} are clipped for display.

after wall removal (Lemma 5). We operationalize this contrast using an hourly volume-weighted average price (VWAP) computed from swap logs. For each swap i in hour h , let p_i denote the alpha price in non-alpha units and v_i the absolute non-alpha amount exchanged; we compute

$$P_h = \frac{\sum_{i \in h} p_i \cdot v_i}{\sum_{i \in h} v_i}.$$

We then construct two metrics from $P(t)$, the VWAP of the hour containing timestamp t . We use *before_price_change* to quantify local drift around wall formation:

$$\text{before_price_change} = \frac{P(t_m) - P(t_m - 1h)}{P(t_m)}.$$

We use *attack_price_change* to quantify displacement after withdrawal relative to the level at t_0 :

$$\begin{aligned} \bar{P}(t_b) &= \frac{P(t_b) + P(t_b + 1h)}{2}, \\ \text{attack_price_change} &= \frac{\bar{P}(t_b) - P(t_0)}{P(t_0)}. \end{aligned}$$

By averaging over the breach hour and the subsequent hour, we capture immediate post-breach movement even when the breach occurs near an hour boundary.

Table 6 in Appendix C reports the binned distributions of both price-displacement metrics.

We expect *before_price_change* to remain near 0 if the wall pins the price, and we expect *attack_price_change* to be large and negative if the attacker dumps toward the floor. The data confirm predictions. Around wall formation, prices remain stable: 16 of 29 episodes fall in the 0%–10% range and only two exceed 20%. During execution, the distribution shifts sharply negative: 20 of 29 episodes fall below -10% , and half of those lie in the -50% to -20% range. Only one episode shows a positive price change, which highlights the one-sided nature of the repricing once the wall is removed.

6.2.4 Liquidity Degradation

Finally, we test the mechanical prerequisite of the dump-to-floor logic: attackers must remove the wall to increase price impact and traverse the curve (Lemma 5). If this mechanism drives the breach, we should observe near-total depletion of one-tick liquidity at t_b . We reconstruct total active one-tick liquidity $L(t)$ by replaying Mint and Burn logs in block-number. We define peak liquidity as $L_{\max} = \max_{t \in [t_m, t_b]} L(t)$ and measure degradation at the breach by

$$\text{drop} = \frac{L_{\max} - L(t_b)}{L_{\max}}.$$

Table 7 in Appendix C reports the binned distribution of liquidity depletion at t_b . Depletion is the norm: eighteen of 29 episodes exhibit a 100% drop (we observe $L(t_b) = 0$), and another seven fall in the 80–99% range. Only four episodes retain more than 20% of peak liquidity at the breach. This near-universal depletion confirms that the breach is not incidental: attackers remove the wall first, leave the attacked tick thin, and then the market reprices sharply as swaps push through the depleted range.

6.3 Participant Behavior and Strategy

We move from system-level market effects to participant-level behavior, using address-level replay to test the participant outcomes implied by Takeaways 2 and 3. The model distinguishes three actors: an attacker who builds and dismantles a one-tick liquidity wall, victim LPs who enter during the stable-yield period, and incentive-driven traders whose activity supplies volume. In this subsection, we first extract observable signals that reveal who controls the wall, then map addresses to theoretical roles using event-anchored replay, and finally quantify the financial outcomes predicted by the model. We conclude by checking robustness to timing conventions and by characterizing how losses and profits concentrate across episodes and addresses.

6.3.1 Robustness to End-Time Choice

Our role inference uses a window end time, but the model’s predictions depend on equilibrium volume and liquidity states rather than a single timestamp. We therefore test whether our conclusions remain stable when we end the episode at t_{oot} (the first pool-internal out-of-tick swap) versus t_b (the hourly breach time used in detection). Table 1 shows that the identified victim set and aggregate loss remain close: using t_{oot} yields 108 positive-loss victims and \$2,809,628.68 total loss, while using t_b yields 99 positive-loss victims and \$2,874,406.50. The victim count differs by 9 addresses, and the aggregate loss differs by only 2.25% relative to the t_b estimate. This indicates that our results do not hinge on a specific end-time convention; the core victim population and loss magnitude persist under both boundaries.

Table 1: Sensitivity of victim identification (n=29).

End Time	Victims	Loss (USD)
t_{oot}	108	2,809,628.68
t_b	99	2,874,406.50
Diff.	+9	-64,777.82
Diff. (% of t_b)	9.09%	-2.25%

6.3.2 Victim Loss Concentration

Having established robustness, we characterize how losses distribute across victims and episodes. Using t_b as the end time, we first measure the unrecovered non-alpha-token value held by LPs who failed to withdraw before the breach. This value is \$10,468,761.31. We then credit the remaining alpha-token value held by these LPs at the breach, which amounts to \$7,594,354.81. The resulting net victim loss is therefore \$2,874,406.50 across 99 positive-loss victim addresses and 13 episodes. This confirms that the liquidity-loading mechanism produces material harm (Theorem 3).

The non-alpha assets involved in the measured losses are USDT, USDC, USD1, and WBNB. For USD aggregation, we value USDT, USDC, and USD1 at \$1 per unit and use a fixed normalization of \$1,000 per WBNB, consistent with the aggregate pricing convention used in our analysis. Under this convention, USDT-denominated losses account for \$2,804,950.86, USDC-denominated losses account for \$55,738.36, USD1-denominated losses account for \$0, and WBNB-denominated losses account for 13.7172812 WBNB, or \$13,717.28 after normalization.

We also observe strong concentration at both the address and episode levels. The single largest victim accounts for 57.1% of total loss and the top five victims account for 84.6%. Similarly, the single largest episode contributes 86.8% of total loss and the top five episodes contribute 97.4%.

The Online Appendix reports detailed rankings of the top victim-loss episodes and addresses. Appendix C reports the corresponding loss-CDF result, showing heavy-tailed concentration at both the victim-address and episode levels.

6.3.3 Attacker Profit Concentration

We next characterize attacker outcomes to test Takeaway 3: once induced activity loads enough liquidity, the resulting surface becomes profitable to exploit. Across 23 episodes with at least one identified attacker address, we identify 149 attacker addresses that earn \$23,884,128.39 in aggregate profit.

We observe heavy-tailed concentration patterns that mirror victim losses. At the episode level, the largest episode accounts for 45.80% of total profit (\$10,938,231.64) and the top five contribute 86.64%, with a median episode profit of \$99,540.52. At the address level, 131 of 149 attacker addresses earn positive profit (median \$21,327.98), and the top

five contribute 43.76%.

The [Online Appendix](#) reports detailed rankings of the top attacker-profit episodes and addresses. [Appendix C](#) reports the corresponding profit-CDF result, showing heavy-tailed concentration at both levels.

6.3.4 Round-Trip Check

The volume-lift result supports [Takeaway 1](#), but it also raises an alternative explanation: the additional volume may come from round-trip trades rather than wall-attracted farmers. We define a round trip as a pair of opposite-direction swaps by the same trader in the same pool, where the trader buys and then sells the alpha token, or sells and then buys it back. If the two legs occur atomically in the same transaction, the trader's price exposure is immediately offset, making the trade largely insensitive to wall depth. In that case, the observed volume would not prove traders being attracted by concentrated liquidity.

We match same-transaction opposite-direction swaps in the Prep windows of the detected episodes. Among 26,514,897 farmer swaps, we identify only 504 atomic round-trip pairs, corresponding to 1,008 swaps, or 0.0038% of farmer swaps. Thus, atomic round trips explain only a negligible share of Prep-window activity. This supports the wall-attraction mechanism: most farmer volume remains exposed to execution costs and is therefore plausibly attracted by the wall's concentrated liquidity.

6.3.5 Comparison with Non-Incentivized Pools

To check whether our detected pattern is specific to incentive-linked pools rather than a generic feature of active CLMM markets, we apply the same detection and validation pipeline to five high-volume PancakeSwap v3 pools outside the Alpha-token incentive set. We denote these pools as P1–P5 and report their addresses in the online appendix.

We do not detect any event satisfying our attack definition in these non-incentivized pools. Four pools show no anomalous build-wall/remove-wall pattern. The remaining pool has large liquidity changes, but they are dispersed across many adjacent ticks rather than concentrated in a one-tick wall, and the maximum post-burn price decline is only 2%.

This comparison does not establish causality, but it suggests that the one-tick wall-and-breach pattern we study is not a generic artifact of high-volume PancakeSwap v3 activity and is more consistent with the incentive-specific mechanism in [Takeaways 1–3](#).

6.3.6 Manual Validation Checks

We conduct two manual checks to validate that the final event set matches our intended attack pattern. First, a one-tick detector could overstate pattern specificity if a broader wall were split across adjacent one-tick positions. We therefore inspect

the target and neighboring ticks, wall-construction/removal transactions, and breach event for each of the 29 validated episodes. We find no validated episode where the wall is constructed through coordinated placement across adjacent one-tick positions.

Second, to distinguish third-party exploitation from token-deployer activity, we compare identified attacker addresses with token deployer addresses. We find no overlap between the attacker and deployer address sets in the validated episodes.

These checks support characterizing the final event set as one-tick attacks. The online appendix reports the per-event validation records.

7 Mitigations

Building on our model and measurements, we discuss four mitigation strategies that target different links in the attack chain. Cap and fee calibration keeps pools away from the profitable attack region; delayed reward settlement weakens the attacker's timing advantage; risk-relevant disclosure reduces victim loading by making manipulation risk visible at LP entry; and anomaly-triggered dynamic caps tighten incentives during suspicious wall-building periods. These strategies are complementary, but each also introduces practical tradeoffs.

7.1 Cap and Fee Calibration

The first strategy is to calibrate the incentive cap and pool fee so that the pool remains outside the profitable attack region. Since [Theorem 6](#) shows that attacks become profitable once $Q \geq Q^*$, platforms can lower Q for high-risk pools or adjust the fee rate f to shift the attacker's payoff below the direct-sale outside option. This reduces cap-funded volume, fee capture, and extraction incentives. Because our measurements show that profits and losses are highly concentrated, platforms could prioritize stricter calibration for pools with concentrated one-tick liquidity, repeated Mint–Burn cycling, or abnormal incentive-driven trading activity. Lowering Q also reduces the maximum subsidized volume a pool can attract, while increasing f raises trading costs and may discourage benign participation. Thus, platforms trade off aggressive growth targets against LP safety.

7.2 Delayed Reward Settlement

The second strategy is to delay the realization of volume-linked rewards, which weakens the attacker's ability to synchronize slow victim accumulation with rapid extraction. Since attackers often keep walls active for days but complete extraction within hours to a day, platforms could settle rewards through vesting, cooldown periods, or time-weighted accrual that discounts short-lived or rapidly reversed volume. Such delay gives LPs more time to observe wall dismantling

near t_0 and react before the breach at t_b ; in model terms, it dampens the yield signal that drives victim entry in Theorem 3 and reduces equilibrium victim liquidity (Lemma 4). Less immediate rewards may also weaken benign participation and reduce the short-term bootstrapping effect of the incentive program.

7.3 Risk-Relevant Information Disclosure

The third strategy is to reduce information asymmetry at LP entry. Since Lemma 4 implies that L_V^* decreases as the effective cost c_V increases, platforms can reduce vulnerable entry by making manipulation risk more salient. They could show a short risk summary before users mint narrow-range liquidity, covering active-tick liquidity concentration, narrow-range churn such as large Burns or repeated Mint–Burn cycling, and whether the pool is in an active incentive window. These signals help LPs distinguish healthy fee yields from potentially manufactured incentive-driven flow; in model terms, they increase the perceived risk component of c_V and reduce the liquidity surface available for extraction. Stronger risk signaling may increase user burden and discourage benign LP participation, reducing the liquidity depth that incentives aim to attract.

7.4 Dynamic Caps with Anomaly Triggers

The fourth strategy is to make the incentive cap Q adaptive to anomaly signals during the preparation window. Platforms could keep generous caps under normal conditions but temporarily tighten caps for pools showing wall-building behavior, such as concentrated one-tick liquidity, large depth increases followed by Burns, or repeated Mint–Burn cycling near the active tick. Since Theorem 1 implies that volume responds predictably to program parameters, lowering Q directly limits manufactured volume and fee capture in at-risk pools. Because preparation usually lasts longer than extraction, anomaly-triggered caps can intervene before the breach rather than after harm occurs. Dynamic caps require continuous monitoring and may introduce false positives, while frequent cap changes can reduce transparency and predictability for benign participants.

8 Limitations

Our study has several limitations. First, our empirical definition requires a highly concentrated liquidity wall. This choice matches the clean one-tick pattern analyzed in our model, but it may undercount less-concentrated variants in which liquidity is spread across several adjacent ticks or a wider price range. Second, our empirical results rely on operational detection choices. Although we report filtering breakdowns and endpoint robustness checks, alternative parameter settings may change some borderline cases. Third, the comparison

with non-incentivized pools is contextual rather than causal, so it cannot fully isolate incentives from other market differences. Fourth, our role labels are inferred from observable on-chain behavior and therefore may not capture private intent, coordination, or off-chain relationships among addresses. Fifth, our economic measurements do not account for all affected stakeholders, such as alpha-token holders or liquidity outside the measured ranges; as a result, they should be interpreted as partial estimates of episode-level impact. Finally, the mitigations are design proposals rather than deployed evaluations, so their effectiveness may differ in broader settings.

These limitations suggest several directions for future work: detecting multi-tick variants, using larger matched or counterfactual samples to better isolate incentive effects, extending role inference with deployer-linkage and address-clustering analysis, decomposing transfers across ticks and participant types, and evaluating mitigations through simulation, historical counterfactuals, or deployment studies.

9 Conclusion

We identify incentive-induced liquidity attacks as an exploitable failure mode in concentrated liquidity market makers. We show that volume-linked incentives can endogenously manufacture order flow, attract liquidity, and create predictable extraction opportunities, even when liquidity providers are rational and correctly price risk. Our equilibrium analysis characterizes a sharp design boundary: incentive programs that push markets into cap-binding regimes face a fundamental tradeoff between volume growth and LP safety. An empirical study on BNB Smart Chain corroborates these predictions, revealing repeated attack episodes with substantial victim losses and consistent execution patterns. Taken together, our results highlight that incentive mechanisms are part of the security perimeter: designs effective at stimulating activity can also be exploited unless incentive strength, observability, and settlement are jointly considered.

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Ethical Considerations

Empirical studies of security and financial systems raise several recurring ethical questions. In our work, these include the scope and timing of the analysis, the potential dual-use implications of describing attack mechanisms, and the handling of sensitive on-chain data. We consider the main stakeholders to include liquidity providers whose capital is exposed to these attacks, incentive-driven traders, protocol and incentive-program designers, auditors and defenders, token projects and platforms that fund incentive programs, and the broader DeFi users who may be affected by degraded market quality. The following discussion explains how these considerations are addressed and what measures are taken to mitigate associated risks.

First, the scope of our study is strictly observational and retrospective. We analyze a class of attacks that have already occurred on public blockchain infrastructure and were completed well before our analysis began. We did not develop, deploy, or assist in any exploit, nor did we interact with attackers or victims. All results are derived from historical transaction traces on BNB Smart Chain (BSC) that are publicly accessible to any observer. As such, our work does not alter system behavior or expose participants to new risks beyond those that have already materialized. For LPs and other market participants, the research process therefore does not create additional exposure; for platforms and protocol designers, it provides a retrospective characterization of risks that are already present in deployed systems.

Second, we recognize the potential for dual use: detailed descriptions of attack mechanics could, in principle, be misused by malicious actors. However, the attacks we study rely on publicly observable mechanisms and do not require privileged access, undisclosed vulnerabilities, or implementation-specific tricks. Moreover, this attack pattern is already known to sophisticated MEV and trading actors, as evidenced by repeated real-world incidents and substantial documented losses. We therefore view publication as beneficial on balance: keeping the mechanism implicit would mainly advantage actors already capable of discovering and exploiting it, while making the pattern explicit helps LPs, auditors, platforms, and incentive-program designers recognize and mitigate the risk. Our contribution is to systematize and formalize these behaviors in a way that makes them visible to protocol designers, auditors, and defenders. We further emphasize that our analytical model is oriented toward identifying the conditions under which such attacks become profitable and toward informing mitigation strategies, rather than providing step-by-step operational guidance. Accordingly, we avoid operational details that would automate exploitation or enable real-time targeting, and we frame the results around detection, risk awareness, and incentive redesign.

Third, we carefully consider data handling and privacy. All analyzed data consists of pseudonymous blockchain ad-

resses and public transaction records. We do not attempt to deanonymize addresses, infer real-world identities, or link on-chain activity to off-chain entities. While we report aggregate statistics on attack frequency, affected liquidity providers, and economic losses, we avoid unnecessary exposure of individual participants. Any address-level information included in the paper or accompanying artifacts is limited to what is required for scientific reproducibility and verification. This limits potential privacy and reputational harms to individual market participants while still allowing independent verification of the empirical claims.

Finally, our decision to conduct and publish this study follows from the same balance of impacts. The research was conducted only on completed, public, historical activity, and the publication is intended to shift knowledge from a small set of sophisticated market actors toward defenders and affected stakeholders. We believe this disclosure is justified because it supports safer incentive design, improves LP risk awareness, and enables future work on detection and mitigation without introducing a new private exploit capability.

Open Science

To support independent verification and follow-on work, we will release a set of research artifacts that enable reproduction of our empirical analyses.

We will make available the scripts used for data collection, preprocessing, and analysis, including code for identifying relevant liquidity pools, reconstructing liquidity distributions across ticks, detecting attack episodes, and computing volume, fee, and loss metrics.

Our queries were executed against the [Allium database](#). All underlying data are publicly available on-chain and do not depend on proprietary access. To ensure reproducibility independent of infrastructure, we also released the SQL queries used in our analysis. These queries specify the data selection and aggregation logic and can be adapted to any comparable blockchain database or archive that supports standard query access to on-chain transaction and event data.

All released artifacts operate solely on publicly available blockchain data. We do not release any scripts that automate live exploitation or interact with protocols in real time, and we limit address-level information to what is necessary for scientific reproducibility. The released scripts, SQL queries, and configuration files are available at <https://zenodo.org/records/20440752>.

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A Proofs

This appendix provides formal proofs for the theorems in Section 4.1.

A.1 Proof of Theorem 1 and Lemma 2

Proof. The trader’s utility from gross volume V is:

$$u_T(V) = \sigma V - V + P_e \cdot x_{\text{out}}(V),$$

where $x_{\text{out}}(V) = L_0 \left(\frac{1}{z_0} - \frac{1}{z_1(V)} \right)$ and $z_1(V) = z_0 + \frac{(1-f)V}{L_0}$ with $L_0 = L_A + L_V$.

Step 1: Concavity. Differentiating:

$$\frac{dx_{\text{out}}}{dV} = \frac{(1-f)L_0}{z_1(V)^2} \cdot \frac{1}{L_0} = \frac{1-f}{z_1(V)^2}.$$

Since $z_1(V)$ is increasing in V , this derivative is positive and decreasing. A function with decreasing derivative is concave, so $x_{\text{out}}(V)$ is concave. Hence $u_T(V)$ is concave.

Step 2: First-order condition. The derivative is:

$$u'_T(V) = \sigma - 1 + P_e \frac{1-f}{z_1(V)^2}.$$

Setting $u'_T(V) = 0$ and solving:

$$z_1^* = \sqrt{\frac{P_e(1-f)}{1-\sigma}}.$$

The implied interior volume is:

$$\tilde{V}(\sigma; L_0) = \frac{L_0}{1-f} \left(\sqrt{\frac{P_e(1-f)}{1-\sigma}} - z_0 \right).$$

Step 3: Participation threshold. At $V = 0$, $u'_T(0) = \sigma - 1 + \frac{(1-f)P_e}{P_0}$. Define $\sigma_0 \equiv 1 - \frac{(1-f)P_e}{P_0}$. The trader participates if and only if $\sigma > \sigma_0$.

Step 4: Piecewise characterization and cap-binding. Under Assumption 2, the tick constraint never binds for any $V \leq Q$ (even when $L_V = 0$). Thus the only potential constraint is the program cap Q , yielding:

$$V^*(\sigma; L_0) = \begin{cases} 0, & \text{if } \sigma \leq \sigma_0, \\ \min\{Q, \tilde{V}(\sigma; L_0)\}, & \text{if } \sigma > \sigma_0. \end{cases}$$

In particular, the cap binds iff $\tilde{V}(\sigma; L_0) \geq Q$. Solving $\tilde{V}(\sigma; L) = Q$ for σ yields the cap-binding threshold

$$\sigma_{\text{cap}}(Q; L) = 1 - \frac{P_e(1-f)}{\left(z_0 + \frac{(1-f)Q}{L}\right)^2}.$$

Assumption 1 imposes $\sigma \geq \sigma_Q \equiv \sigma_{\text{cap}}(Q; L_A)$, which is sufficient for $V^* = Q$ for all $L_V \geq 0$. $\square$

A.2 Proof of Theorem 3 and Lemma 4

Proof. Under Assumptions 1 and 2, equilibrium volume is $V^* = Q$ independent of L_V . The victim's expected payoff is:

$$\Pi_V(L_V) = fQ \cdot \frac{L_V}{L_A + L_V} - c_V L_V,$$

where $c_V > 0$ captures opportunity cost plus anticipated im-permanent loss.

Step 1: First derivative.

$$\Pi'_V(L_V) = fQ \cdot \frac{L_A}{(L_A + L_V)^2} - c_V.$$

Step 2: Second derivative.

$$\Pi''_V(L_V) = -2fQ \cdot \frac{L_A}{(L_A + L_V)^3} < 0 \quad \text{for } fQ > 0, L_A > 0.$$

The objective is strictly concave.

Step 3: Interior FOC.

Setting $\Pi'_V(L_V) = 0$:

$$fQ \cdot \frac{L_A}{(L_A + L_V)^2} = c_V \iff (L_A + L_V)^2 = \frac{fQL_A}{c_V}.$$

Solving:

$$L_V^{\text{int}}(Q) = \sqrt{\frac{fQL_A}{c_V}} - L_A.$$

Step 4: Boundary conditions.

Imposing $L_V \geq 0$:

$$L_V^*(Q) = \max \left\{ 0, \sqrt{\frac{fQL_A}{c_V}} - L_A \right\}.$$

Step 5: Comparative statics.

- (i) $\frac{\partial L_V^*}{\partial Q} = \frac{1}{2} \sqrt{\frac{fL_A}{c_V Q}} > 0$ at interior: L_V^* is increasing in Q .
- (ii) $\frac{\partial L_V^*}{\partial c_V} = -\frac{1}{2} \sqrt{\frac{fQL_A}{c_V^3}} < 0$ at interior: L_V^* is decreasing in c_V .
- (iii) At interior, $L_A + L_V^* = \sqrt{fQL_A/c_V}$.

□

Remark (Myopic Variant). If victims do not anticipate the attack, they use $c_V^{\text{myopic}} < c_V^{\text{rational}}$. By comparative static (ii), this yields $L_V^{\text{myopic}} > L_V^{\text{rational}}$, amplifying victim loading.

A.3 Proof of Lemma 5 (Dump-to-Floor)

Proof. Let $\Lambda(z)$ denote the liquidity available at square-root price z . Under our one-tick abstraction, $\Lambda(z) = L_V$ for $z \in [z_L, z_U]$ and $\Lambda(z) = 0$ otherwise.

The dump payoff is:

$$\Pi_{\text{dump}}(z_2; z_1) = U_{\text{dump}}(z_2; z_1) - c_A X_{\text{dump}}(z_2; z_1),$$

where:

$$U_{\text{dump}}(z_2; z_1) = \int_{z_2}^{z_1} \Lambda(z) dz,$$

$$X_{\text{dump}}(z_2; z_1) = \frac{1}{1-f} \int_{z_2}^{z_1} \frac{\Lambda(z)}{z^2} dz.$$

Step 1: Differentiate with respect to z_2 .

By Leibniz's rule:

$$\frac{\partial \Pi_{\text{dump}}}{\partial z_2} = -\Lambda(z_2) \left(1 - \frac{c_A}{(1-f)z_2^2} \right).$$

Step 2: Sign analysis.

For $\Lambda(z_2) > 0$:

- If $z_2^2 > \frac{c_A}{1-f}$: $\frac{\partial \Pi_{\text{dump}}}{\partial z_2} < 0$, so decreasing z_2 increases pay-off.
- If $z_2^2 < \frac{c_A}{1-f}$: $\frac{\partial \Pi_{\text{dump}}}{\partial z_2} > 0$, so increasing z_2 increases pay-off.

Step 3: Marginal-cost stopping point.

$$z^\dagger = \sqrt{\frac{c_A}{1-f}}.$$

Step 4: Inventory constraint. Let $\bar{A}_4 = \bar{A} - X_{\text{sold},3}$ denote remaining inventory. The dump requires $X_{\text{dump}} \leq \bar{A}_4$, implying $z_2 \geq z_{\bar{A}}$.

Step 5: Constrained optimum. The endpoint must satisfy $z_2 \leq z_1$ (feasibility). Thus:

$$z_2^* = \max \{ z_{\text{floor}}, z_{\bar{A}}, z^\dagger \} \quad \text{subject to } z_2^* \leq z_1.$$

Step 6: One-tick baseline. By the liquidity function definition, $z_{\text{floor}} = z_L$. By Assumption 2, the Stage-3 endpoint $z_1(Q) \leq z_U$, so the dump starts within the tick. Moreover, for typical parameters $z^\dagger < z_1(Q)$, ensuring feasibility. □

A.4 Proof of Theorem 6 (Attack Feasibility)

Proof. Assume Assumptions 1–2 hold, so that equilibrium trading volume is $V^* = Q$ independent of $L_V \geq 0$.

Step 1: Victim entry and the Stage-3 endpoint are increasing in Q . By Theorem 3,

$$L_V^*(Q) = \max \left\{ 0, \sqrt{\frac{fQL_A}{c_V}} - L_A \right\}$$

is continuous and nondecreasing in Q . Let $L_0(Q) \equiv L_A + L_V^*(Q)$. Stage-3 trading moves the square-root price to

$$z_1(Q) = z_0 + \frac{(1-f)Q}{L_0(Q)}.$$

If $L_V^*(Q) = 0$, then $z_1(Q) = z_0 + \frac{(1-f)Q}{L_A}$ is strictly increasing in Q . If $L_V^*(Q) > 0$, then $L_0(Q) = \sqrt{fQL_A/c_V}$ and

$$z_1(Q) = z_0 + (1-f)\sqrt{\frac{c_V}{fL_A}} \sqrt{Q},$$

which is also strictly increasing in Q . Hence $z_1(Q)$ is strictly increasing on the feasible domain.

Step 2: The fee component is strictly increasing in Q . The attacker's fee income at cap-binding volume is

$$F(Q) \equiv fQ \cdot \frac{L_A}{L_0(Q)}.$$

If $L_V^*(Q) = 0$, then $F(Q) = fQ$. If $L_V^*(Q) > 0$, then $L_0(Q) = \sqrt{fQL_A/c_V}$ and thus

$$F(Q) = fQ \cdot \frac{L_A}{\sqrt{fQL_A/c_V}} = \sqrt{fL_A c_V} \sqrt{Q}.$$

In both cases $F(Q)$ is continuous and strictly increasing in Q .

Step 3: The optimal dump value is nondecreasing in Q . Let $\pi_{\text{dump}}(x; z_1, L_V)$ denote the net payoff from dumping x units of token0 through victim-only liquidity L_V starting from square-root price z_1 :

$$\pi_{\text{dump}}(x; z_1, L_V) \equiv U(x; z_1, L_V) - c_A x,$$

where (within a tick) the post-dump price $z_2(x)$ solves $\frac{1}{z_2(x)} = \frac{1}{z_1} + \frac{(1-f)x}{L_V}$ and the token1 proceeds are $U(x; z_1, L_V) = L_V(z_1 - z_2(x))$. For any fixed $x > 0$, U (and hence π_{dump}) is increasing in both z_1 and L_V under the one-tick formulas.

Define the stage-4 value function

$$D(Q) \equiv \max_{0 \leq x \leq \bar{A}} \pi_{\text{dump}}(x; z_1(Q), L_V^*(Q)),$$

where $\bar{A}$ is the attacker's dump inventory bound. Since $z_1(Q)$ is increasing and $L_V^*(Q)$ is nondecreasing, $D(Q)$ is nondecreasing in Q (pointwise maximum of nondecreasing functions). Continuity of $D(Q)$ follows from Berge's maximum theorem (continuous objective, compact action set).

Table 2: Filtering breakdown of the detection pipeline.

Stage	Description	Events	Pools
S0	Alpha-matched CLMM pools	N/A	3,249
S1	Valid Mint/Burn traces	N/A	209
S2	One-tick anomaly groups	34,819	123
S3	Build/remove wall pairs	5,359	90
S4	$\geq 10\%$ post-removal drop	128	76
S5	Pool-internal breach	94	53
S6	Replay and micro-checks	30	18
S7	Downstream reconstruction	29	18

Step 4: Monotonicity and threshold. Collecting the Q -independent terms into a constant (e.g., the capital cost $\kappa(L_A)$), the attack payoff can be written as

$$\Pi_A^Q = F(Q) + D(Q) - \kappa(L_A).$$

Since $F(Q)$ is strictly increasing and $D(Q)$ is nondecreasing, Π_A^Q is continuous and strictly increasing in Q .

Define

$$Q^* \equiv \inf\{Q \in [0, V_{\max}^{\text{tick}}(L_A)] : \Pi_A^Q \geq \Pi_{\text{direct}}\}.$$

If the set is empty then attacks are never profitable on the feasible range. Otherwise, continuity implies $\Pi_A^{Q^*} = \Pi_{\text{direct}}$, and strict monotonicity implies uniqueness. Finally, by monotonicity, attacks are profitable if and only if $Q \geq Q^*$. $\square$

A.5 Proof of Corollary 7 (Volume–Security Tradeoff)

Proof. By Theorem 1, $V^*(\sigma)$ is nondecreasing in σ , so for any volume target $\underline{V}$, there exists a minimum subsidy $\sigma_{\underline{V}} = \inf\{\sigma : V^*(\sigma) \geq \underline{V}\}$.

By Theorem 6, attacks are profitable when $\sigma \geq \sigma_Q$ and $Q \geq Q^*$. The safe region is: $\sigma < \sigma_Q$ or $Q < Q^*$.

The optimal safe subsidy solves $\sigma^{\text{opt}} = \min\{\sigma : V^*(\sigma) \geq \underline{V}\}$ subject to safety. If $\underline{V}$ requires $\sigma \geq \sigma_Q$ with $Q \geq Q^*$, the platform faces a binding tradeoff. $\square$

B Detection Pipeline Breakdown

Table 2 summarizes the complete filtering pipeline corresponding to the four-stage methodology in Section 5: (1) Alpha-token pool scoping (§5.1), (2) one-tick anomaly-group construction (§5.2 Candidate Identification), (3) price-drop and pool-internal validation (§5.2 Validation), and (4) replay-based micro validation and downstream reconstruction (§5.3).

C Additional Tables and Figures

This appendix provides supplementary tables and figures that support the empirical results reported in Section 6.

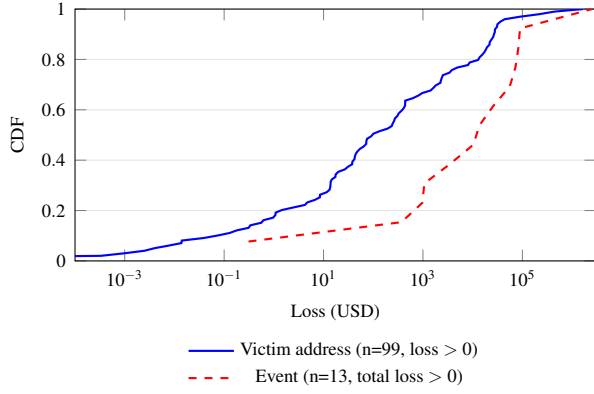


Figure 3: CDF of clipped MTM victim loss in USD. The solid line is positive-loss victim addresses, and the dashed line is positive-loss events aggregated by event.

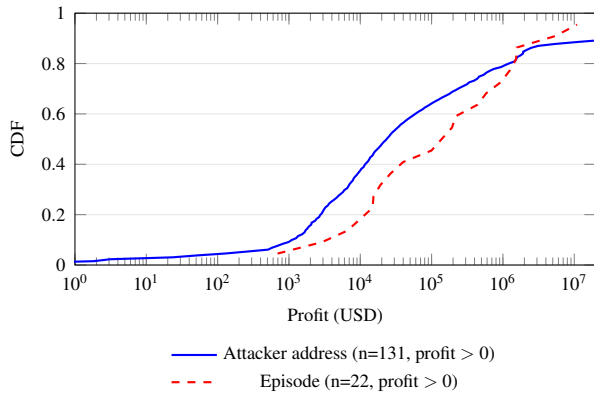


Figure 4: CDF of attacker profit measured in the crash window $[t_0, t_b]$. We plot only positive-profit entries to support log-scaled x-axis.

Figure 3 plots the distribution of measured victim losses at two aggregation levels: positive-loss victim addresses and positive-loss events. The address-level curve shows that most victims incur relatively small losses, while the event-level curve shows that aggregate losses are concentrated in a small number of episodes. Figure 4 reports the analogous distribution for attacker profits, separately for positive-profit attacker addresses and positive-profit episodes. The heavy right tail in both curves supports the concentration patterns discussed in Section 6.3.

Detailed top-5 rankings of victim losses and attacker profits are reported in the [Online Appendix](#). Figures 3 and 4 in Appendix C report the corresponding CDF results, which show heavy-tailed concentration at the victim-address, episode, attacker-address, and attacker-episode levels.

Table 3: Binned distributions of temporal gaps (n=29).

Bucket	Δ_{build}	Δ_{extract}	Δ_{oot}
same_time	3	6	1
$\leq 30\text{min}$	0	8	3
30min–1h	0	0	0
1h–6h	5	4	6
6h–12h	3	2	3
12h–24h	2	3	6
1d–3d	4	3	7
$> 3\text{d}$	12	3	3

Table 4: Binned distributions for volume lift metrics (n=29).

Bucket	$Lift_{\text{prep}}$	$Lift_{\text{atk}}$	$Lift_{\text{post}}$
$< 1\times$	9	12	23
$1\text{--}1.5\times$	11	9	1
$1.5\text{--}2\times$	4	3	1
$2\text{--}5\times$	2	1	2
$> 5\times$	3	4	2

Table 5: Binned distributions at t_b (n=29).

Bucket (drop ratio)	count
0–20%	0
20–40%	0
40–60%	2
60–80%	2
80–99%	7
100%	18

Table 6: Binned distributions for price changes (n=29).

Bucket	before_price_change	attack_price_change
$< -50\%$	1	0
-50% to -20%	0	10
-20% to -10%	1	10
-10% to 0%	7	8
0% to 10%	16	1
10% to 20%	2	0
20% to 50%	2	0
$> 50\%$	0	0

Table 7: Binned distributions at t_b (n=29).

Bucket (drop ratio)	count
0–20%	0
20–40%	0
40–60%	2
60–80%	2
80–99%	7
100%	18